

ROCK HILL, SC, USA

WMO: 723117

Lat: 34.987N

Lon: 81.058W

Elev: 204

StdP: 98.90

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 53871

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.5	-4.4	-16.4	0.9	-1.4	-13.7	1.2	-0.7	8.8	11.1	8.0	10.7	1.5	340	0.384

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	34.8	23.6	33.6	23.5	32.5	23.3	25.5	31.4	25.0	30.8	24.5	30.0	2.9	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.0	19.3	27.4	23.4	18.6	26.8	22.9	18.1	26.4	79.0	31.5	76.7	30.9	74.8	30.0	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.6	6.6	5.5	DB	-10.3	36.7	2.2	1.6	-11.8	37.8	-13.1	38.8	-14.3	39.7	-15.9	40.9	(4)
(5)			WB	-11.1	26.3	2.2	0.6	-12.7	26.8	-13.9	27.1	-15.2	27.5	-16.8	27.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.4	5.5	7.3	11.4	16.4	20.7	25.1	26.5	25.9	22.9	16.6	10.8	7.4	(6)
(7)		DBStd	8.49	5.53	4.95	5.07	4.17	3.53	2.36	1.89	2.23	2.97	4.29	4.52	4.82	(7)
(8)		HDD10.0	473	162	104	44	4	0	0	0	0	0	3	44	112	(8)
(9)		HDD18.3	1687	399	309	219	84	20	0	0	4	83	229	339	(9)	(9)
(10)		CDD10.0	2823	22	29	89	195	333	453	512	494	388	209	67	32	(10)
(11)		CDD18.3	996	0	1	6	26	95	203	253	236	143	30	2	1	(11)
(12)		CDH23.3	9012	1	6	61	296	823	1913	2421	2128	1102	242	17	2	(12)
(13)	CDH26.7	3726	0	0	7	64	263	842	1131	947	418	52	1	0	(13)	
(14)	Wind	WSAvg	2.2	2.5	2.6	2.7	2.8	2.3	2.1	2.0	1.7	1.9	2.1	2.2	(14)	
(15)	Precipitation	PrecAvg	1191	108	101	123	82	86	109	103	107	106	88	84	89	(15)
(16)		PrecMax	1719	205	228	260	203	225	265	251	351	309	299	211	252	(16)
(17)		PrecMin	743	11	22	19	17	13	12	11	13	0	0	12	13	(17)
(18)		PrecStd	200	50	50	56	49	48	59	59	69	77	68	43	46	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.3	24.3	27.7	30.4	33.0	35.9	36.3	36.5	34.2	30.7	25.9	22.7	(19)
(20)			MCWB	16.7	17.1	17.6	18.9	22.3	23.3	24.3	23.4	21.9	22.6	18.2	17.6	(20)
(21)		2%	DB	19.0	20.9	25.1	28.6	31.2	34.1	34.8	34.3	32.6	27.8	22.9	20.1	(21)
(22)			MCWB	14.9	14.3	16.1	18.7	21.4	23.3	24.1	23.7	21.9	20.2	16.4	16.4	(22)
(23)		5%	DB	17.1	18.6	22.8	26.9	29.6	32.7	33.4	32.9	31.1	26.2	21.1	18.0	(23)
(24)			MCWB	13.4	13.1	15.0	17.5	20.7	23.2	24.0	23.8	22.0	19.0	16.0	15.1	(24)
(25)	10%	DB	14.6	16.3	20.3	24.8	27.9	31.3	32.2	31.6	29.1	24.0	18.9	16.1	(25)	
(26)		MCWB	11.2	11.4	13.6	16.8	20.2	22.7	23.8	23.5	21.6	18.6	14.6	13.1	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.0	18.6	18.8	21.6	23.9	25.4	26.3	26.1	25.2	24.3	20.7	19.6	(27)
(28)			MCDWB	20.4	22.4	24.3	26.0	29.3	32.0	32.3	31.5	29.7	27.0	22.1	20.9	(28)
(29)		2%	WB	16.3	16.7	17.7	20.4	22.7	24.7	25.6	25.4	24.2	22.7	18.8	17.6	(29)
(30)			MCDWB	18.3	19.7	22.4	25.3	28.0	31.2	31.8	31.0	28.5	25.5	20.9	19.1	(30)
(31)		5%	WB	14.4	14.6	16.5	19.3	21.9	24.1	25.0	24.9	23.5	21.2	17.4	15.8	(31)
(32)			MCDWB	16.5	17.5	21.0	24.2	27.4	30.3	31.2	30.3	27.4	23.9	19.6	17.6	(32)
(33)	10%	WB	11.8	12.3	15.3	18.1	21.2	23.5	24.5	24.3	22.7	19.8	15.7	13.5	(33)	
(34)		MCDWB	14.0	15.0	18.8	22.7	26.2	29.3	30.2	29.4	26.5	23.1	18.4	15.2	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.4	11.7	12.6	13.2	11.8	11.3	10.9	10.5	10.7	12.1	12.7	11.2	(35)
(36)			MCDBR	13.5	14.4	14.9	15.3	13.5	13.2	12.5	12.4	13.5	13.8	14.5	12.5	(36)
(37)			MCWBR	9.2	8.8	7.6	6.7	5.3	4.4	4.0	3.9	4.7	6.5	8.6	8.8	(37)
(38)		5% WB	MCDBR	11.3	11.9	12.4	12.3	11.1	11.4	10.9	10.5	10.1	10.3	10.5	10.1	(38)
(39)			MCWBR	9.0	8.6	7.4	6.1	4.7	4.3	4.0	3.9	4.1	5.7	7.8	8.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.315	0.324	0.361	0.405	0.461	0.487	0.538	0.518	0.432	0.385	0.334	0.315	(40)	
(41)		taud	2.498	2.468	2.394	2.264	2.149	2.116	1.997	2.039	2.265	2.373	2.476	2.530	(41)	
(42)		Ebn at Noon	892	922	910	881	833	808	763	771	826	839	859	870	(42)	
(43)		Edh at Noon	83	96	112	134	153	158	177	167	127	105	85	77	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.66	3.31	4.38	5.47	6.06	6.42	6.01	5.50	4.69	3.87	2.94	2.29	(44)	
(45)		RadStd	0.15	0.38	0.32	0.42	0.49	0.46	0.39	0.26	0.41	0.47	0.24	0.21	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (67 neighbors)		+0.53	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+170	+117

Nomenclature: See separate page